

Enterprise IT Infrastructure Security Assessment & Risk Analysis

Overview

Conducted a structured academic case study assessing the security posture of a telecom-scale IT infrastructure. Focused on identifying vulnerabilities, analyzing risks, and proposing mitigation strategies aligned with industry frameworks.

Scope

- Security Architecture Mapping
- Policy Review
- Vulnerability Assessment
- Risk Analysis
- Compliance Evaluation
- Executive Reporting

Security Architecture

Analyzed core infrastructure components including networks, systems, applications, and access controls.

Policy Review

Evaluated policies against ISO 27001 and NIST CSF. Identified gaps in updates, training, and third-party risk management.

Key Vulnerabilities

- Unpatched legacy systems
- Weak authentication mechanisms
- SQL injection exposure
- Unencrypted internal communication

Risk Analysis

Developed a prioritization model based on impact and likelihood. Critical risks included legacy systems and web vulnerabilities.

Recommendations

- Replace legacy systems
- Strengthen authentication (MFA)
- Conduct regular assessments
- Implement DLP
- Improve third-party risk management

Compliance

ISO 27001, NIST CSF, Saudi NCA, PDPL

Note

Academic case study based on simulated assessment findings.